

India releases list of 30 critical minerals crucial for 'economic development, national security'

Barring a few, India 100% dependent on imports for most of the listed minerals, which find use in high-tech electronics, telecommunications, transport and defence.

Annexure II

Usage and availability of Identified Critical Minerals

S. No.	Critical Minerals	Value Chain	Major Applications	Examples of Specific Products	Indian Availability	Remarks
1	Antimony	Advanced manufacturing	Flame retardants, Lead-acid batteries, Lead alloys, Plastics (catalysts and stabilisers), Glass and ceramics	Manufacture of chemicals and chemical products, electrical equipment, fabricated metal products, except machinery and equipment	No proved reserves and only inferred reserves are available in Lahul & Spiti districts of Himachal Pradesh	- The metal is obtained commonly as a by-product in lead-zinc-silver smelting by Hindustan Zinc Limited - The bulk of secondary antimony has been recovered at secondary lead smelters as antimonial lead, most of which was generated and then consumed by the Lead- acid Battery Industry - Possibilities exists to recover antimony from lamp phosphor waste (which is considered as a source of rare-earth elements)
2	Beryllium	Advanced manufacturing inputs and materials	Automotive components: Transport and Defence- Manufacturin g of Machinery., Electronic and telecommunic ations equipment's	Manufacture of computer, electronic and optical products	Not available	Current requirement is made through imports
3	Bismuth	Advanced manufacturing	Chemicals, Pharmaceutic als, Casting of Iron	Manufacture of chemicals and chemical products	Not available	Current requirement is made through imports
4	Cadmium	Advanced manufacturing	Batteries, Pigments, Coatings	Manufacture of electrical equipment & chemicals and chemical products, solar cells, electroplating, and silver soldering	Cadmium is recovered as a by-product during zinc smelting and refining. The total annual installed capacity for recovering cadmium was 913 tpy of which HZL accounted for 833 tpy capacity	

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5	Cobalt	Clean technologies	Electric Vehicle, Batteries, corrosion resistant alloys, aerospace applications, Pigments and Dyes	Battery electrodes; metal alloys; turbine engine components, automobile airbags; catalysts in the petroleum and chemical industries; drying agents for paints, varnishes, and inks; magnets, organic and inorganic chemical compounds, textiles, paper, leather and like industries, high speed specialized cutting tools, glass, and ceramics industry	Not available	Current requirement is made through imports
6	Copper	Clean technologies and advanced manufacturing	Electrical and electronics products, Electrical Wiring, Solar Panel, Automotive industry	Power transmission lines, electrical building wiring, vehicle wiring, telecommunication wiring, electronic components, fabricated metal products	The current production of the Copper concentrate is able to meet only 4 % of the requirement of Copper Smelter/refineries necessitating a huge number of imports. To fill the gap in demand and supply of Copper, the existing mines have to scale up the production as per the mining plan/ EC limits as well as new mines have to be developed	
7	Gallium	Information and communications	Semiconductors, Integrated Circuits, LEDs	Electronic circuit boards, LED devices, semiconductors, specialized thermometers, barometric sensors, solar panels, blue-ray technology, pharmaceuticals	Gallium is recovered as a by-product while producing alumina. Two plants, namely, HINDALCO at Renukoot, Uttar Pradesh and NALCO Damanjodi alumina refinery, Odisha, had recovered gallium in the past. Needs to be revisited	
8	Germanium	Information and communications, clean technologies, and advanced manufacturing	Optical fibres, satellites, solar cells	Fibre-optic communication networks, camera and microscope lenses, infrared night vision systems, polymerization catalysts	Not available	Current requirement is made through imports
9	Graphite	Clean technologies	Batteries, Lubricants, fuel cells for EVs, Electric Vehicle	Metal foundry lubricants, vehicle brake linings, metal casting wear, crucibles, rechargeable battery anodes, EV fuel cells, electrical motor components, frictionless materials, pencils	9 million tonnes reserves exist	Production reported from 12 mines
10	Hafnium	Advanced manufacturing	Superalloy, Catalyst precursor, Semiconductors, Oxide for Optical, Nuclear reactors	Manufacture of basic metals, computer, electronic and optical products, neutron absorption in control rods in nuclear power plants, used in iron, titanium, niobium, tantalum, and other metal alloy, liquid rocket thruster nozzle	Normally, all zirconium compounds contain between 1.4% and 3% hafnium. IREL & KMML are involved in production of zircon	

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11	Indium	Advanced manufacturing	Electronics (Laptops, LED Monitors/TVs, Smartphones), and semi conductors	Electronics, electrical equipment	Not available	Current requirement is made through imports
12	Lithium	Clean technologies and defence & security technologies	Electric Vehicle, Batteries, glassware, ceramics, fuel manufacturing, Lubricant	Rechargeable batteries (phones, computers, cameras, and EVs); hydrogen fuel storage; metal alloys (military ballistic armour; aircraft, bicycle, and train components); specialized glass and ceramics; drying and air conditioning systems, steam or other vapour generating boilers, pharmaceuticals, medicinal chemical and botanical products, refined petroleum products	Not available	Current requirement is made through imports
13	Molybdenum	Advanced manufacturing	Steel alloys, Pigment and Dyes, Catalyst, Electrical and Electronic	Engineering steels, Foundries, Nickel alloys, Lubricants, Field of Medical	Mineable reserves are available in Harur of Tamil Nadu State	Mining Blocks needs to be auctioned to avoid import route
14	Niobium	Clean technologies and advanced manufacturing	Construction, transportation	Metal alloys (steel), jet engines, rockets, construction beams, building girders, oil rigs and pipelines, superconducting magnets, MRI scanners, NMR equipment, eyeglasses, titanium niobium oxide anode materials, nuclear power plants, production of electronic components	Not available	Current requirement is made through imports
15	Nickel	Clean technologies and advanced manufacturing, defence & security technologies	Stainless steel, solar panels, batteries, aerospace, defence applications and Electric Vehicle	Metal alloys (steel, superalloys, non-ferrous alloys), jet and combustion engine components, rechargeable batteries (phones, computers, EVs), industrial manufacturing machines, construction beams, anti-corrosive pipes, cookware, medical implants, power plant components	Vedanta has a nickel and cobalt plant at Goa named NICOMET. Nickel sulphate and Cobalt sulphate production is 7500TPA	

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16	PGE	Advanced manufacturing	Auto catalyst, Jewellery, medicine, electronic equipment used by military	Used in making optical fibers, LCDs, turbine blades, spark plugs, fabrication of silicon in aerospace, guided missile systems, catalytic converters in reducing toxic emissions, hybrid integrated circuits, medicine (pacemakers, chemotherapy) and jewellery	About 14.2 tonnes of metal content of PGM are located in Niligiri, Boulasahi and Sukinda areas in Odisha and remaining 1.5 tonnes of metal content of PGM in Hanumalpur area in Shivamogga schist belt of Karnataka	
17	Phosphorous	Advanced manufacturing	Mineral fertilizer	Utility as a nutrient provider, Detergents, chemicals, food additives, Animal Feed, rust removers, corrosion preventers	Huge reserves/resources are available in Rajasthan, Jharkhand and Madhya Pradesh. There are around 7 reporting mines	
18	Potash	Advanced manufacturing	Chemical Fertilizers, Water softener	Industrial chemicals, animal feed, soap making, food production, water softening, road de-icing, pH adjustment, explosives, pharmaceuticals, and glassmaking	Major part of resources are located in Nagaur district of Rajasthan, followed by Panna district, Madhya Pradesh and the balance in Sonbhadra & Chitrakoot districts, Uttar Pradesh	
19	REE	Zero-emission vehicles, defence & security technologies	Permanent magnets for electricity generators and motors, catalyst, polishing, Batteries, Electronics, Defence technologies, wind energy sector, aviation, and Space	Laptops, LED Monitors/TVs, Smartphones, EV drive trains, wind turbines, aircraft components, vehicle components, speakers, steel manufacturing, battery anodes, chemical catalysts, glass manufacturing, specialized glass lenses, hybrid and tactical wheeled vehicles, armoured vehicles, night vision goggles, communication, and navigation systems	Resource estimate of monazite from beach sand in India is 11.93 Mt having 55%-65% of rare earth oxides	Met through domestic production by IREL and through imports
20	Rhenium	Advanced manufacturing	Super-alloys, aerospace and machinery uses, Catalysts in petroleum industry	Manufacture of coke and refined petroleum products	Not available	Current requirement is made through imports

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21	Selenium	Advanced manufacturing inputs and materials	Electrolytic, Manganese, Glass, pigments	Pigments for ceramics, paint and plastics, useful in photocells, solar cells and photocopiers, used in anti-dandruff shampoos, used as an additive to make stainless steel	No production.	
22	Silicon	Information and communications, advanced manufacturing	Semiconductors, electronics, and transport equipment, Paints, Aluminum alloys	Electronics, computer processors and photovoltaics, steel manufacturing, glass, and glass products	India is reporting production of 59000 metric tonnes of silicon and ranks 12 th in production as per 2022 data	
23	Strontium	Advanced Manufacturing	Alloys of aluminium, Pigments and Fillers, Glass, Magnets, Pyrotechnic application	Casting of non-ferrous metals, parts and accessories of three wheelers and motorcycles	Not available	Current requirement is made through imports
24	Tantalum	Clean technologies and advanced manufacturing	Capacitors, Superalloys, Carbides, Medical technology	Manufacture of machinery and equipment, used in Nickel-based super alloys for aircrafts, gas turbine components, missiles and radio communication systems, preparation of advanced airframes in Defence sector, digital camera, mobile phones, high temperature furnace parts, alloys for air and land-based turbines	Not available	Current requirement is made through imports
25	Tellurium	Clean technologies	Solar power, thermoelectric devices, Rubber vulcanising	Metal alloys (copper and steel), solar cells, semiconductors, CDs/DVDs, vulcanized rubber, chemical catalysts in oil refining	No production	
26	Tin	Advanced manufacturing	Aerospace, construction, home decor, electronics, jewellery and telecommunications	Lead acid batteries, Tinplate, Metal Packaging, decorative elements for homes, countertops, Solders, chemicals	Produced in the form of concentrates and metal in Chhattisgarh by CMDC & Precision Minerals & Smelting in Dantewada	From domestic production and through partial imports

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27	Titanium	Clean technologies and advanced manufacturing, defence & security technologies	Aerospace and defence applications, chemicals and petro-chemicals, Pigments, Polymers	Colour pigments in paint, plastics, and paper; metal alloys (aluminium, steel, molybdenum); aircraft; spacecraft; missiles and rockets; non-corrosive pipes; ship and submarine hulls; medical implants; sunscreen; specialty Li-ion battery anode materials	Titanium Sponge Plant of India is located at Kerala Minerals and Metals Ltd (KMML), Chavara, Kollam district of Kerala	Major resources / reserves are available in the coastal districts of Tamilnadu, Andhra Pradesh, Odisha, Kerala, Gujarat & Maharashtra
28	Tungsten	Advanced manufacturing	Mill and cutting tools, Mining and construction tools, Catalysts and pigments, Aeronautics and energy uses, tungsten carbide	Consumed for production of hard materials like high penetration alloy for weaponry, rockets, missiles, cutting tools (tungsten carbide), filament wires, electrodes and super alloys, oil and gas drilling	Not available	Current requirement is made through imports
29	Vanadium	Clean technologies and advanced manufacturing	Alloys, batteries	Metal alloys (steel), military armour plating, vehicle axles, piston rods and crankshafts, vanadium redox flow batteries, nuclear reactor components, manufacturing of superconducting magnets, pigments for ceramics and glass	Total estimated reserves / resources of vanadium ore as on 1.4.2015 are placed at 24.63 million tonnes with an estimated V2O5 content of 64,594 tonnes	R&D on opportunity for production from mine tailings of bauxite has to be strengthened in addition to opening up of new mines
30	Zircon	Clean technologies and advanced manufacturing	High value chemical manufacturing and electronics sector,	Good nuclear fuel-rod cladding metal, useful in nose cones, re-entry vehicles and coatings in jet engines, used in heat shield in space shuttles, as catalytic fuel converters as well as refractories	IREL & KMML are involved in production	Obtained as a by-product during the beneficiation of heavy mineral sands in Manavalakurichi in Tamil Nadu, Chavara

The following value chains have the highest potential:

- Clean technologies, which include zero-emission vehicles (ZEV), wind turbines, solar panels, advanced batteries, hydrogen fuel cells, small modular reactors.
- Information and communication technologies, including semiconductors.
- Advanced manufacturing inputs and materials, such as defence applications, permanent magnets, ceramics, high value-added metals, electronic materials, composites, polymers, and biomaterials.